

# Supply, habitat and the price of liquidity in less-liquid sovereign bond markets.

Evidence from Poland.

**Marcin Dec**

Kozminski University & FAME|GRAPE, Warsaw

Borsa Istanbul Review, 2026 · open access · doi:10.1016/j.bir.2026.100904



NCN Preludium UMO-2020/37/N/HS4/02202

# | What actually moves bond liquidity?

## THE QUESTION

---

- In a less-liquid sovereign market, liquidity is not a fixed property of a bond. It is the joint output of issue size, supply, holder structure, the trading-venue rulebook and the macro regime.
- This paper treats liquidity as the **dependent variable**, not as an input to curve fitting.
- Five hypotheses, four data layers, twenty-one years of public data on the Polish sovereign curve.

# | The setting: less-liquid markets (LLM)

## THE SETTING

---

- The middle tier of sovereign curves: visible enough that every dealer prices liquidity into a quote, transient enough that tick-data estimators built for the deepest curves do not transfer.
- Daily fixings and monthly turnover are the highest-frequency public record.
- Poland is the best-instrumented member of the cohort. Comparable LLMs: Brazil, Chile, Czechia, Hungary, India, Indonesia, Israel, Korea, Malaysia, Mexico, Singapore, South Africa, Thailand, Türkiye.

# Five hypotheses

## WHAT WE TEST

---

- **H1. Issue size narrows everything.** Larger outstanding  $\Rightarrow$  tighter spreads, lower price impact.
- **H2. Stocks, not flow.** Residual-maturity composition prices into liquidity, monthly issuance flow does not.
- **H3. Foreign retreat hits Amihud, not the quote.** Buy-side foreign holders exit; volume-scaled measures move, the venue-capped quote does not.
- **H4. MPC transmission is regime-dependent.** Pre- and post-COVID monetary days price into liquidity through different channels.
- **H5. Auctions leave an inventory imprint.** Primary Dealers absorb at concession and unwind into the secondary fixing.

# Finding 1: issue size is the most robust bond-level driver

H1 SUPPORTED

---

—3.5 bp

bond-month coefficient on  $\log(\text{outstanding})$  for the quoted spread,  $p < 0.01$ .  
Same sign on the composite z-score and on the zero-trading-days share.

Survives bond fixed effects, macro and regime controls,  
across all six liquidity estimators.

## Finding 2: supply matters through stocks, not flow

H2 SUPPORTED

---

- Monthly gross issuance does not move segment liquidity once stock measures are in.
- Trailing-twelve-month rolling issuance and the log of segment stock load monotonically; the residual-maturity share loads positively on Amihud and on zero-trading days.

**A well-supplied segment with an even maturity ladder trades tighter than a thin segment with a clumpy ladder.**

# Finding 3: the venue cap decomposes BAS from Amihud

H3 SUPPORTED

The buy-side foreign share of zloty-denominated sovereign debt fell from  $\sim 33\%$  to under 15% over 2019–2024.

## Quoted spread (BAS)



*capped at 30–90 ticks by the Annex G venue rules*

## Amihud price impact



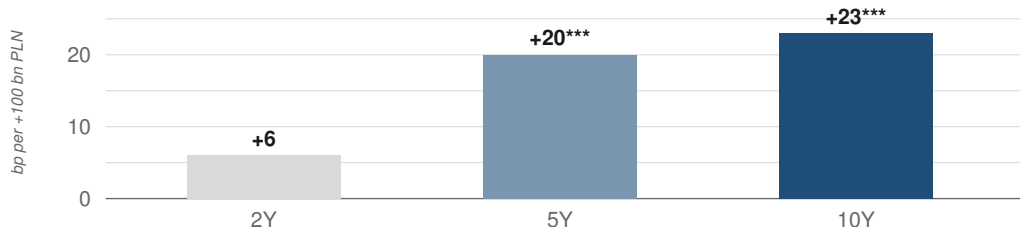
*free to widen:  $+0.29^{***}$  on the foreign share*

The same buy-side retreat is invisible in the quote and strongly visible in the price-impact metric. Carry both measures: the decomposition is the information.

## Finding 4: supply prices into the term premium with a maturity gradient

HABITAT SIGNATURE

Coefficient on the total State Treasury debt level (per +100 bn PLN) in monthly regressions of ACM term premia:



Increasing in tenor: a preferred-habitat signature. Plus about +8 bp at 10Y per percentage point of debt-to-GDP.

## Finding 5: the supply wedge is macro-relevant

### AGGREGATE PRICE TAG

---

Reprice the Polish fixed-cash-flow universe under the current curve, the BRW risk-neutral curve, and the curve net of the supply-attributable term-premium wedge from the 2019–2025 debt rise:

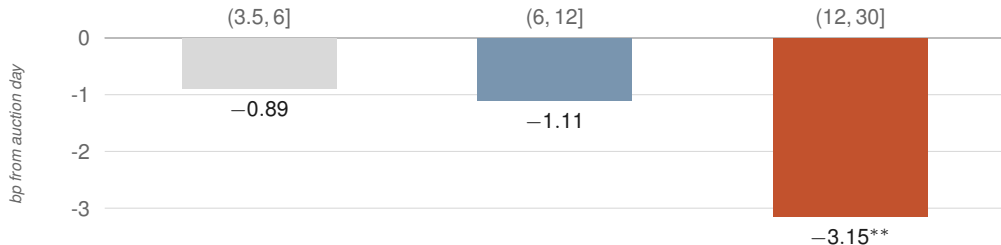
**2.4%**

of the present value of the 951 bn PLN fixed-cash-flow universe is the mark-to-market difference attributable to the supply-driven term premium accumulated over the 2019–2025 debt expansion. About 23 bn PLN.

## Finding 6: primary auctions print a small inventory premium

H5 IN THE CREDIBLE WINDOW

Mean change in segment-mean fixing yield from the auction-day level at  $k = +5$  business days, pre-2022 sub-sample:



Yields drift down post-event and the magnitude rises with maturity: the textbook auction-cycle reversion of Lou–Yan–Chang (2013), now documented on an LLM.

# | How to read the result

## WHAT IT MEANS

---

- **For practitioners.** Quote-based and volume-scaled liquidity metrics are not interchangeable when the venue caps the quote. Carry both: the decomposition is the information.
- **For policymakers.** The supply gradient on the term premium is real: a 100 bn PLN debt rise prices into the ten-year term premium by about 23 bp through the level channel, plus another 8 bp per percentage point of debt-to-GDP.
- **For researchers.** Treat sovereign liquidity on a less-liquid market as an endogenous object driven by stocks, holders and venue rules, not as a fixed bond attribute.

# Methodology in one slide

ONE PANEL, FOUR LAYERS

---

- **Data.** Daily BondSpot fixings and turnover on 91 PLN-denominated sovereign bonds, 2005–2026, plus Ministry of Finance holdings and a macro panel.
- **Measures.** Six low-frequency estimators: quoted bid-ask, Amihud, Roll, Bao–Pan–Wang gamma, Corwin–Schultz, composite z-score, plus the zero-trading-days share.
- **Econometrics.** Bond-month and segment-month panels with fixed effects and Driscoll–Kraay errors; MPC event study with a pre/post-COVID split; auction event study on the yield path; aggregate monthly regressions of BAS and ACM term premia on supply.
- **Valuation.** A three-curve present-value scenario pricing the supply-attributable wedge on the fixed-cash-flow universe.

# | The results live on, daily

YIELDCARTOGRAPHY.COM

---

- The paper's six measures and the composite are recomputed on the full BondSpot panel with every site build, normalisation constants frozen on the paper sample so the live series stays comparable with the published one: [yieldcartography.com/liquidity](https://yieldcartography.com/liquidity).
- ACM and BRW term premia behind the supply regressions update daily on the [/term-premia/](#) tab, and the cockpit carries the headline tiles.
- All underlying CSVs are downloadable from the dashboards; the paper is open access.

# Thank you.

Marcin Dec

Kozminski University & FAME|GRAPE, Warsaw

mdec@kozminski.edu.pl · ORCID 0000-0002-6220-8267

Dec, M. (2026). Supply, habitat and the price of liquidity in less-liquid sovereign bond markets. Evidence from Poland. Borsa Istanbul Review. doi:10.1016/j.bir.2026.100904 · open access

Research supported by the National Science Centre, Poland · Preludium UMO-2020/37/N/HS4/02202



[yieldcartography.com/slides/liquidity/](https://yieldcartography.com/slides/liquidity/)